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AI Engineer

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SE3 Labs
AI and Computer Vision Team
Munich

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Application for Data and ML Engineer

Dear SE3 Labs team,

I am writing to apply for the Data and ML Engineer position at SE3 Labs. I have a Master's in Artificial Intelligence and three years of experience building and deploying AI applications, and the part of that work I enjoy most is getting messy real-world data into a shape a model can learn from.

At Covestro I trained models on real process data from prepolymer synthesis batches. I took raw sensor time series (temperature, reaction time, process variables), built lag features and rolling windows for Random Forest and XGBoost, fed the raw sequences into an LSTM, and benchmarked all of them against traditional empirical curve-fitting. The result helped spot batches likely to miss their target specification early, and a Streamlit tool let the chemists try parameter changes before running a physical batch. That project taught me how much model quality depends on how the data is cleaned and prepared. Later, in my Master's thesis, I benchmarked multilingual embedding models on over one million chunks of Covestro data and fine-tuned models for retrieval.

At Kniit I am the sole developer of a knowledge-extraction pipeline for the German energy sector, from ingestion and chunking to indexing, and I report directly to the CTO. Evaluation drives everything there: I measured every change against a ground truth derived from SAP system data and raised precision and recall from 33% and 45% to 93% and 90%. Domain experts review and approve every change to the shared knowledge store through a web app I built.

To be clear about the gaps: I have not yet worked with 3D vision, UAV recordings or field data collection, and I write Python rather than C++. My Bachelor's is in mechanical engineering and my publication is on autonomous indoor vehicles, so I am at home around hardware, and I am glad to go out to test sites and help collect the data myself.

SE3 Labs comes out of Daniel Cremers's computer vision group at TUM, and I like that the role is about turning that research into datasets and models the whole company can reuse. Your systems have to keep working when GPS and other sensors drop out, and that comes down to how well the training data covers those conditions. Building that data foundation is the work I want to do.

I will be based in Munich, hold an EU Blue Card and can start on 15 November 2026. Thank you for considering my application.

Kind regards,

Chandu Prasad Bhairapu